

Thevenin's Theorem

Definition

Thevenin's Theorem states that any complex, linear two-terminal circuit can be replaced by an equivalent circuit consisting of a single voltage source (V_{th}) in series with a resistance (R_{th}) connected to a load.

$$I = V_{th} / (R_{th} + R_L)$$

This equation allows direct calculation of current through the load resistor. Instead of solving the entire circuit repeatedly, the simplified model provides instant results for different load values.

What is a Linear Circuit?

A **linear circuit** is one in which the relationship between voltage and current is proportional and follows Ohm's Law.

$$V \propto I \rightarrow V = IR$$

In linear circuits:

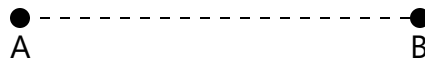
- Doubling the voltage doubles the current
- Parameters such as resistance remain constant
- The principle of superposition is valid

Examples of linear components include resistors, inductors, and capacitors (under normal conditions). Non-linear elements such as diodes and transistors do not follow this proportional behavior.

Step-by-Step Thevenin Method

Step 1: Disconnect the Load

Remove the load resistor and identify the two terminals. This isolates the portion of the circuit to be simplified.



Step 2: Find R_{th}

Deactivate all independent sources. Voltage sources become short circuits and current sources become open circuits. Then calculate equivalent resistance.

$$R_{th} = R_{eq}$$

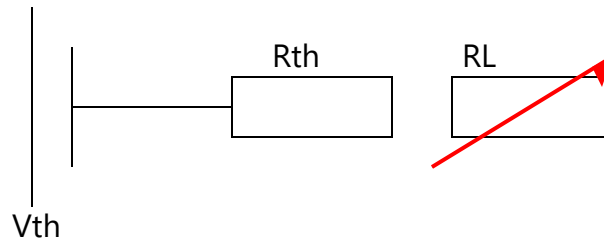
Step 3: Find V_{th}

Turn on all sources and calculate the open-circuit voltage across the terminals.

$$V_{th} = V_{oc}$$

Step 4: Construct and Solve Equivalent Circuit

After obtaining V_{th} and R_{th} , the circuit is reconstructed. The load resistor is reconnected to the simplified circuit.



The arrow across the resistor represents a **variable resistor**, meaning its resistance can change. This allows analysis of how current and voltage respond to different load values.

Methods to Solve the Equivalent Circuit

1. Nodal Analysis

This method applies Kirchhoff's Current Law (KCL) at circuit nodes. It is most useful when the circuit has multiple nodes and fewer loops.

Use when: Many parallel branches or complex node connections.

2. Mesh Analysis

Mesh analysis uses Kirchhoff's Voltage Law (KVL) to analyze loops in a circuit.

Use when: Circuit has fewer loops and is planar.

3. Source Transformation

A voltage source in series with a resistor can be converted into an equivalent current source in parallel with a resistor, and vice versa.

$V \text{ source} \leftrightarrow I \text{ source} \quad (V = IR)$

Use when: Simplifying mixed-source circuits.

4. Ohm's Law

For simple circuits, current can be directly calculated using:

$$I = V / R$$

Use when: Circuit is already simplified.

5. Superposition Theorem

Each independent source is analyzed separately and results are summed.

Use when: Multiple independent sources are present.

Practical Applications

Thevenin's theorem is widely used in engineering to simplify circuit analysis. It is especially valuable in power systems, where sections of a grid are replaced with equivalent circuits for efficient load analysis.

In electronics, it is used in designing circuits that must operate under varying load conditions, such as amplifiers and sensors. It is also used in battery modeling to predict performance under different loads.

Overall, the theorem reduces complexity and provides a clear understanding of how circuits interact with connected components.